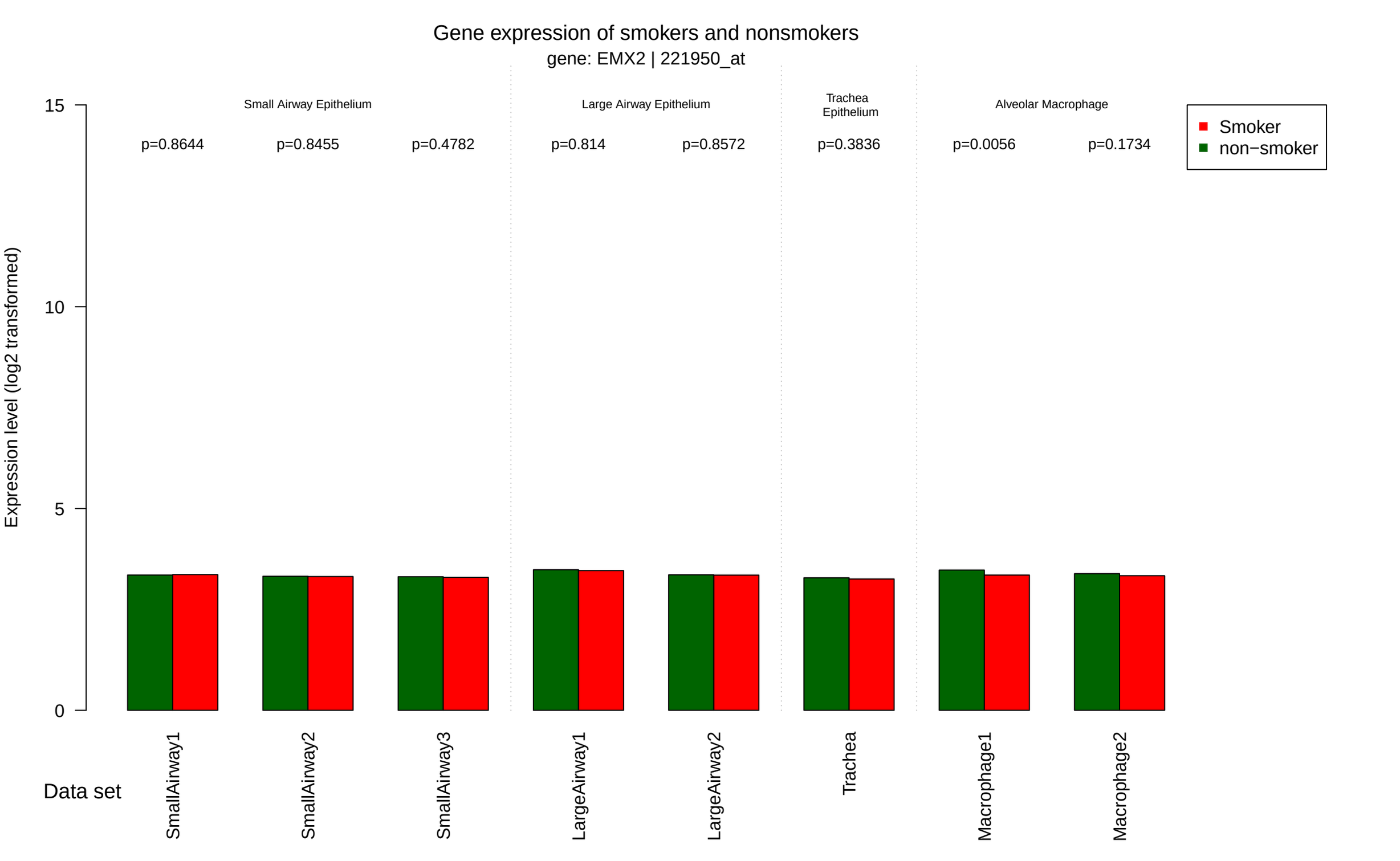
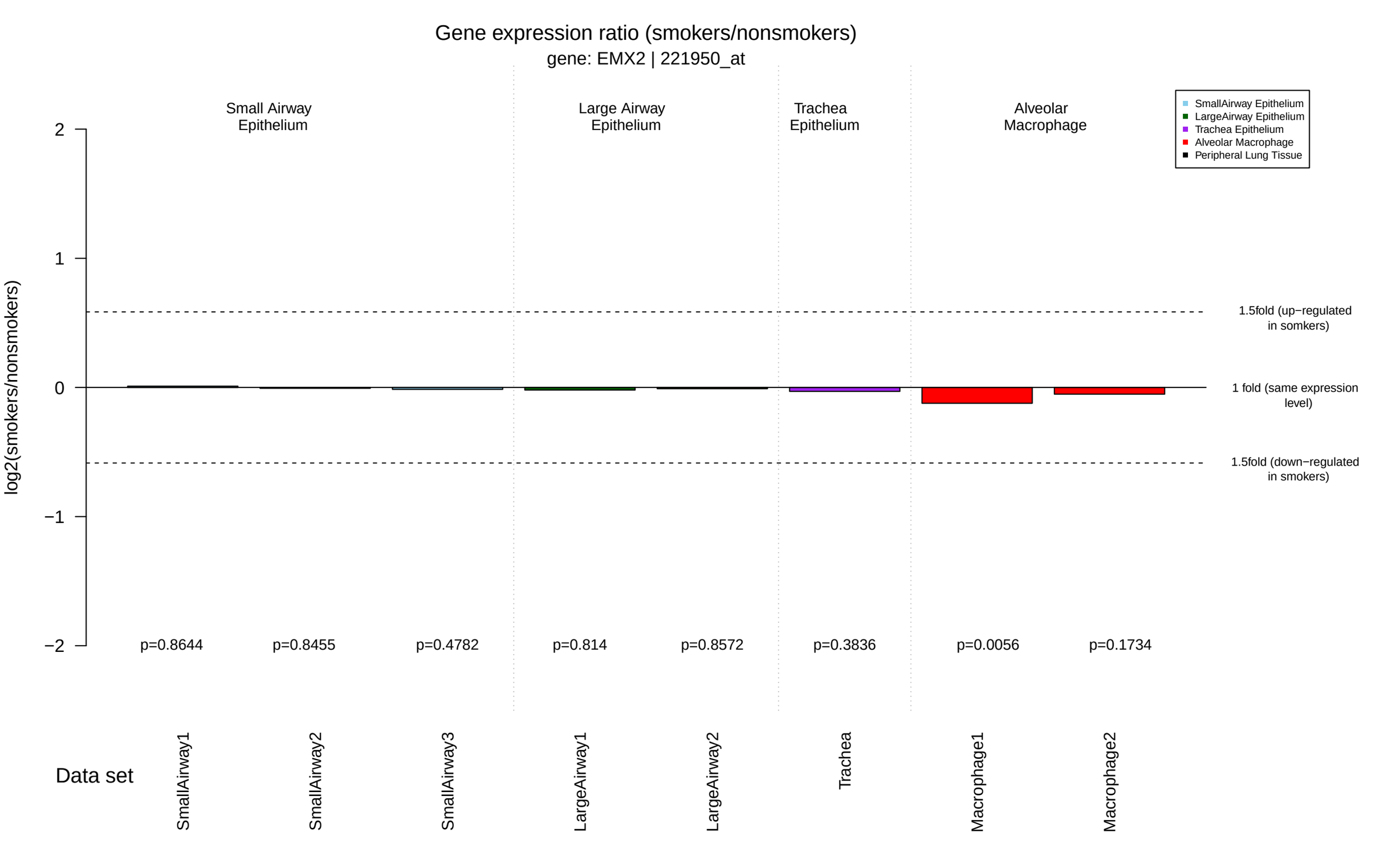
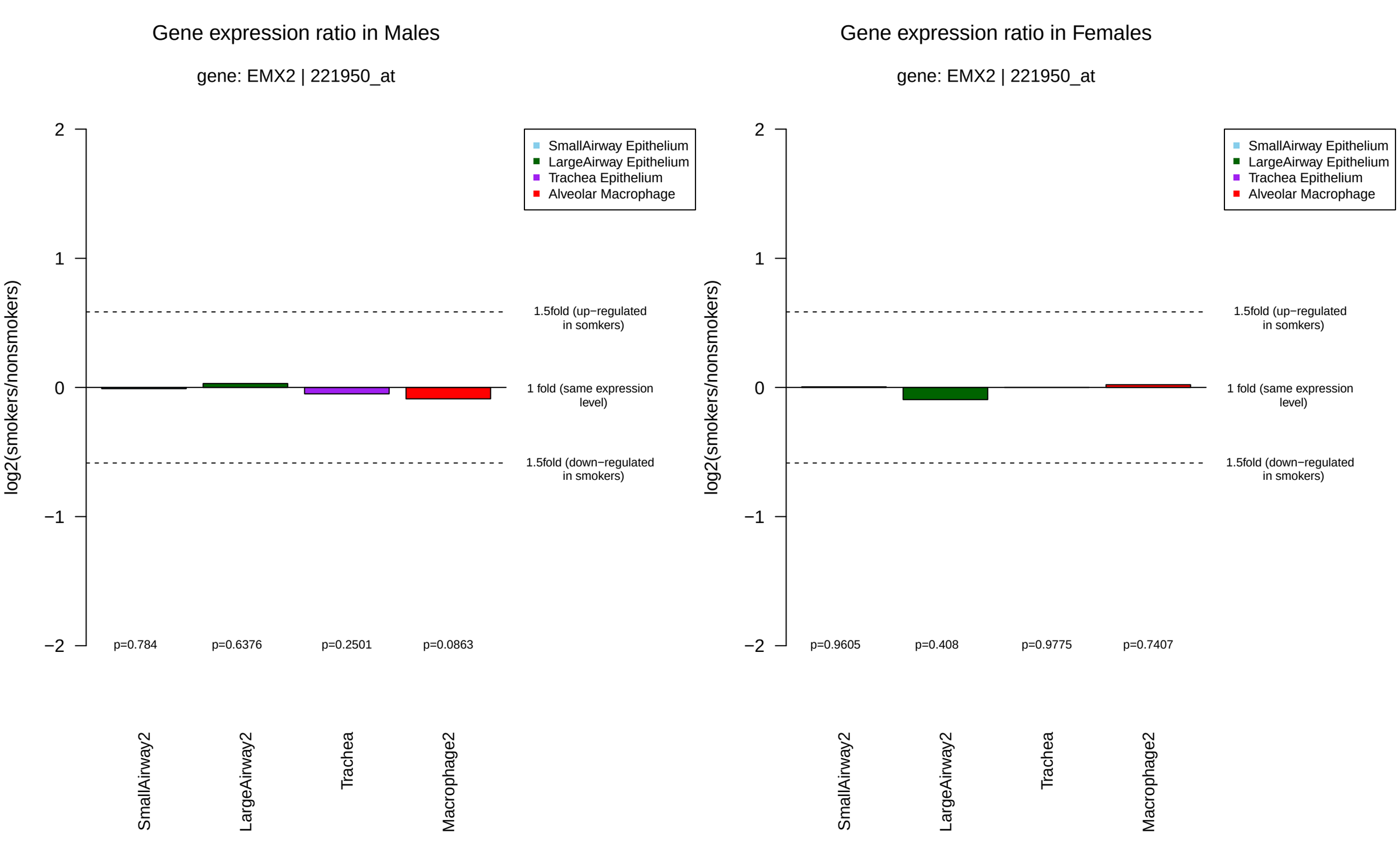
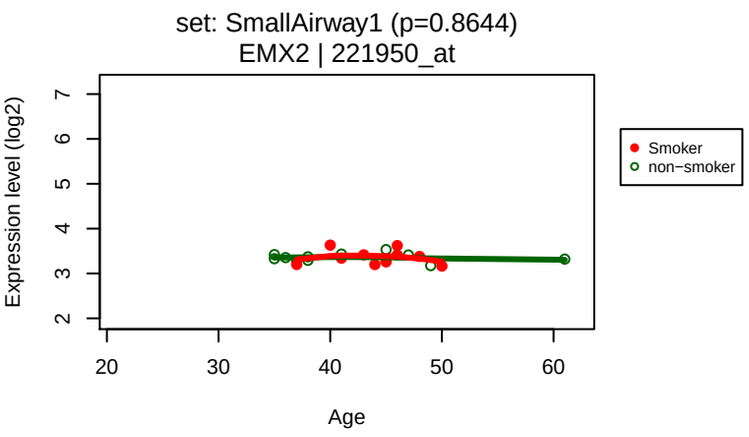
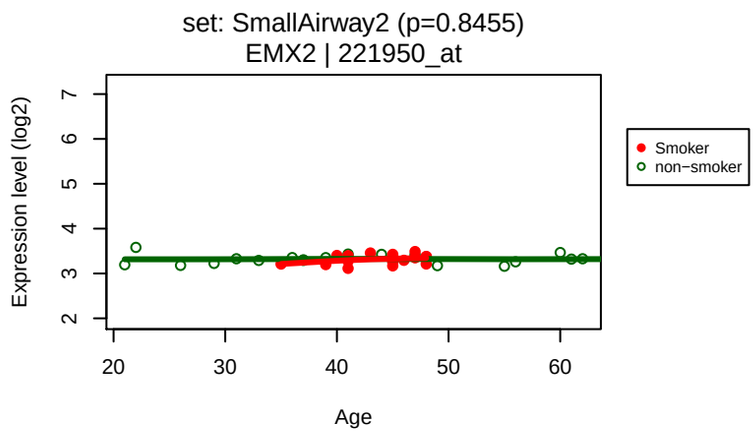
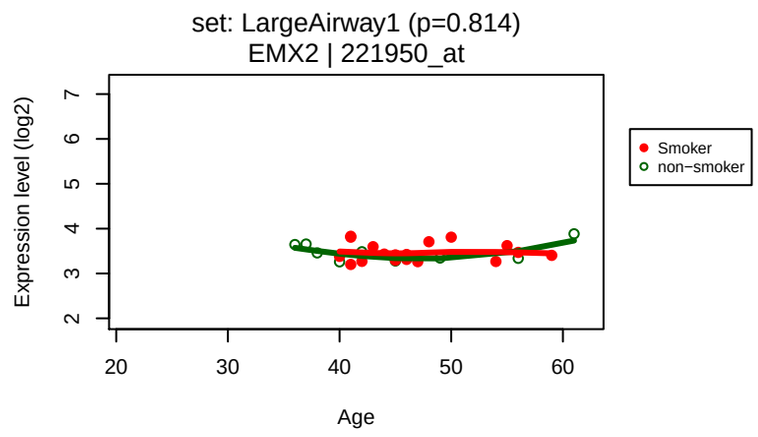

Gene expression of smokers and nonsmokers
gene: EMX2 | 221950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8644
p=0.8455
p=0.4782
p=0.814
p=0.8572
p=0.3836
p=0.0056
p=0.1734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

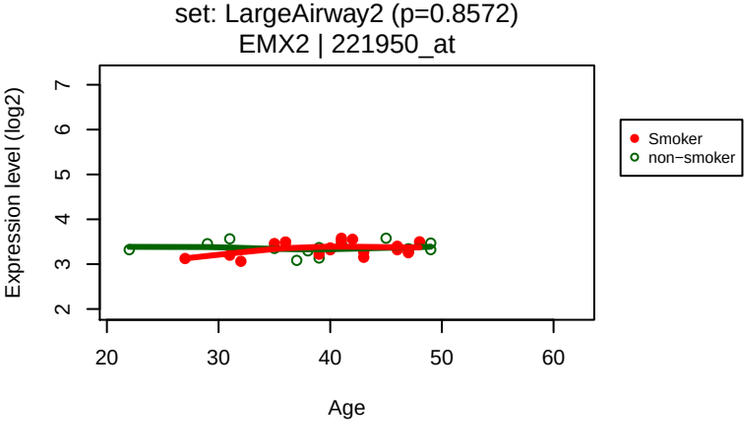

Gene expression ratio (smokers/nonsmokers)
gene: EMX2 | 221950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8644
p=0.8455
p=0.4782
p=0.814
p=0.8572
p=0.3836
p=0.0056
p=0.1734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

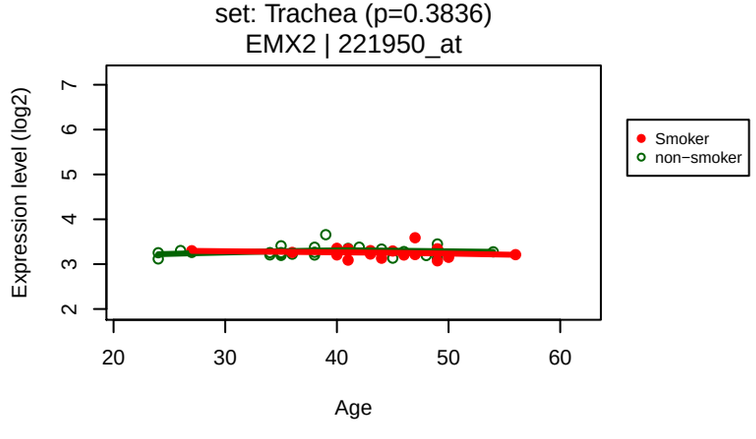

Gene expression ratio in Males
gene: EMX2 | 221950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.784
p=0.6376
p=0.2501
p=0.0863
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EMX2 | 221950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9605
p=0.408
p=0.9775
p=0.7407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

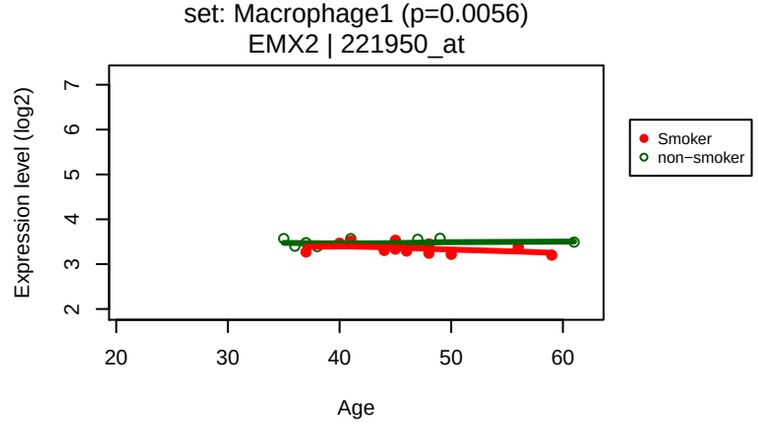

set: SmallAirway1 (p=0.8644)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker

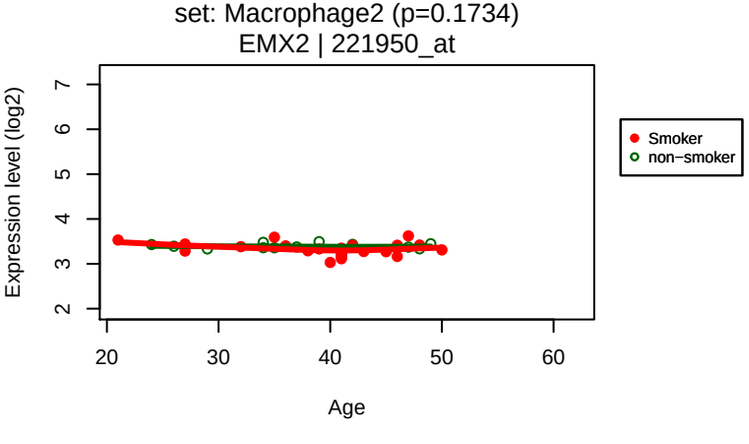

set: SmallAirway2 (p=0.8455)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker

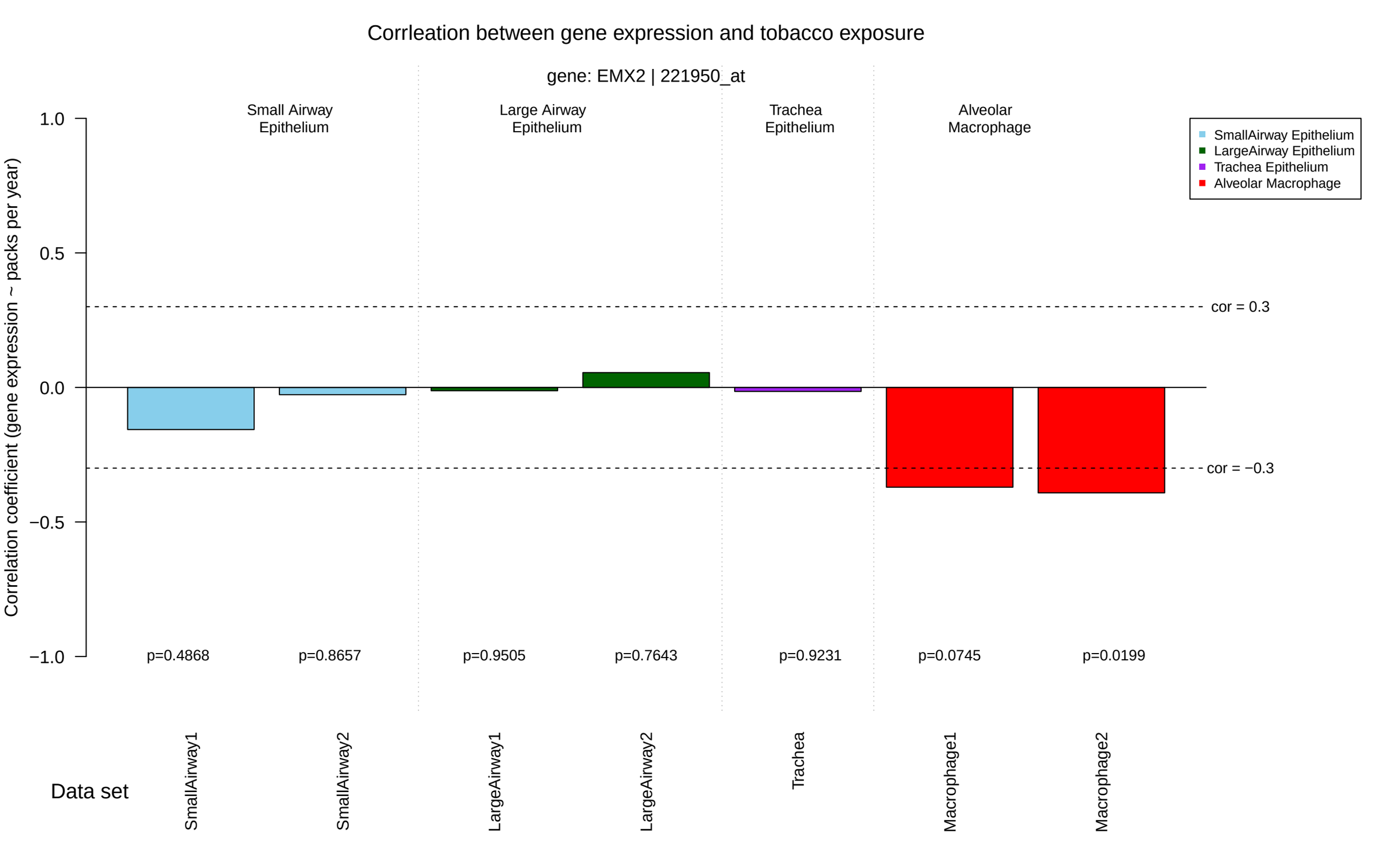

set: LargeAirway1 (p=0.814)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8572)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3836)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0056)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1734)
EMX2 | 221950_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EMX2 | 221950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4868
p=0.8657
p=0.9505
p=0.7643
p=0.9231
p=0.0745
p=0.0199
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2